

Variations in materials, manufacturing processes and testing result in motor-to-motor efficiency variations for a given motor design; the full-load efficiency for a large population of motors of a single design is not a unique value but rather a band of efficiency. Therefore, the energy efficiency limits given in this standard are nominal.

5.3 Classification and marking

5.3.1 General

The designation of the energy efficiency class consists of the letters “IE” (short for International Energy efficiency class), directly followed by a numeral representing the classification according to Table 2.

5.3.2 Efficiency classification

Table 2 – IE efficiency classification

Designation	Definition
IE1	Motors with a rated full-load efficiency equal to or exceeding the limits listed in 5.4.1.
IE2	Motors with a rated full-load efficiency equal to or exceeding the limits listed in 5.4.2.
IE3	Motors with a rated full-load efficiency equal to or exceeding the limits listed in 5.4.3.
IE4	Motors with a rated full-load efficiency equal to or exceeding the limits listed in 5.4.4.
IE5	Envisaged for a future edition of this standard. See Annex A.

5.3.3 Motors below IE1 efficiency

Some motors have rated efficiencies below the limits given in Tables 3 and 4. No marking of these motors shall be required.

5.3.4 Marking

The rated efficiency and the IE code shall be durably marked on the rating plate, for example “IE2 – 84,0 %”.

5.4 Nominal limits for efficiency classes IE1, IE2, IE3 and IE4

NOTE Shaded areas in the tables indicate changes compared to the previous edition of this standard.

5.4.1 Nominal efficiency limits for IE1

Table 3 – Nominal efficiency limits (%) for 50 Hz IE1

P_N kW	Number of poles/synchronous speed min ⁻¹			
	2/3000	4/1500	6/1000	8/750
0,12	45,0	50,0	38,3	31,0
0,18	52,8	57,0	45,5	38,0
0,20	54,6	58,5	47,6	39,7
0,25	58,2	61,5	52,1	43,4
0,37	63,9	66,0	59,7	49,7
0,40	64,9	66,8	61,1	50,9
0,55	69,0	70,0	65,8	56,1
0,75	72,1	72,1	70,0	61,2
1,1	75,0	75,0	72,9	66,5
1,5	77,2	77,2	75,2	70,2
2,2	79,7	79,7	77,7	74,2
3	81,5	81,5	79,7	77,0
4	83,1	83,1	81,4	79,2
5,5	84,7	84,7	83,1	81,4
7,5	86,0	86,0	84,7	83,1
11	87,6	87,6	86,4	85,0
15	88,7	88,7	87,7	86,2
18,5	89,3	89,3	88,6	86,9
22	89,9	89,9	89,2	87,4
30	90,7	90,7	90,2	88,3
37	91,2	91,2	90,8	88,8
45	91,7	91,7	91,4	89,2
55	92,1	92,1	91,9	89,7
75	92,7	92,7	92,6	90,3
90	93,0	93,0	92,9	90,7
110	93,3	93,3	93,3	91,1
132	93,5	93,5	93,5	91,5
160	93,8	93,8	93,8	91,9
200	94,0	94,0	94,0	92,5
250	94,0	94,0	94,0	92,5
315	94,0	94,0	94,0	92,5
355	94,0	94,0	94,0	92,5
400	94,0	94,0	94,0	92,5
450	94,0	94,0	94,0	92,5
500 up to 1 000	94,0	94,0	94,0	92,5

Table 4 – Nominal efficiency limits (%) for 60 Hz IE1

P_N kW	Number of poles/synchronous speed min ⁻¹			
	2/3600	4/1800	6/1200	8/900
0,12	57,5	62,0	48,0	36,0
0,18	62,0	66,0	52,5	40,0
0,25	64,0	68,0	57,5	50,5
0,37	70,0	70,0	62,0	57,5
0,55	72,0	74,0	66,0	59,5
0,75	74,0	77,0	72,0	64,0
1,1	78,5	79,0	75,0	73,5
1,5	81,0	81,5	77,0	77,0
2,2	81,5	83,0	78,5	78,0
3,7	84,5	85,0	83,5	80,0
5,5	86,0	87,0	85,0	84,0
7,5	87,5	87,5	86,0	85,0
11	87,5	88,5	89,0	87,5
15	88,5	89,5	89,5	88,5
18,5	89,5	90,5	90,2	88,5
22	89,5	91,0	91,0	90,2
30	90,2	91,7	91,7	90,2
37	91,5	92,4	91,7	91,0
45	91,7	93,0	91,7	91,0
55	92,4	93,0	92,1	91,5
75	93,0	93,2	93,0	92,0
90	93,0	93,2	93,0	92,5
110	93,0	93,5	94,1	92,5
150 up to 1 000	94,1	94,5	94,1	92,5

5.4.2 Nominal efficiency limits for IE2 (see Tables 5 and 6)

Table 5 – Nominal efficiency limits (%) for 50 Hz IE2

P_N kW	Number of poles/synchronous speed min ⁻¹			
	2/3000	4/1500	6/1000	8/750
0,12	53,6	59,1	50,6	39,8
0,18	60,4	64,7	56,6	45,9
0,20	61,9	65,9	58,2	47,4
0,25	64,8	68,5	61,6	50,6
0,37	69,5	72,7	67,6	56,1
0,40	70,4	73,5	68,8	57,2
0,55	74,1	77,1	73,1	61,7
0,75	77,4	79,6	75,9	66,2
1,1	79,6	81,4	78,1	70,8
1,5	81,3	82,8	79,8	74,1
2,2	83,2	84,3	81,8	77,6
3	84,6	85,5	83,3	80,0
4	85,8	86,6	84,6	81,9
5,5	87,0	87,7	86,0	83,8
7,5	88,1	88,7	87,2	85,3
11	89,4	89,8	88,7	86,9
15	90,3	90,6	89,7	88,0
18,5	90,9	91,2	90,4	88,6
22	91,3	91,6	90,9	89,1
30	92,0	92,3	91,7	89,8
37	92,5	92,7	92,2	90,3
45	92,9	93,1	92,7	90,7
55	93,2	93,5	93,1	91,0
75	93,8	94,0	93,7	91,6
90	94,1	94,2	94,0	91,9
110	94,3	94,5	94,3	92,3
132	94,6	94,7	94,6	92,6
160	94,8	94,9	94,8	93,0
200 up to 1 000	95,0	95,1	95,0	93,5

Table 6 – Nominal efficiency limits (%) for 60 Hz IE2

P_N kW	Number of poles/synchronous speed min ⁻¹			
	2/3600	4/1800	6/1200	8/900
0,12	59,5	64,0	50,5	40,0
0,18	64,0	68,0	55,0	46,0
0,25	68,0	70,0	59,5	52,0
0,37	72,0	72,0	64,0	58,0
0,55	74,0	75,5	68,0	62,0
0,75	75,5	78,0	73,0	66,0
1,1	82,5	84,0	85,5	75,5
1,5	84,0	84,0	86,5	82,5
2,2	85,5	87,5	87,5	84,0
3,7	87,5	87,5	87,5	85,5
5,5	88,5	89,5	89,5	85,5
7,5	89,5	89,5	89,5	88,5
11	90,2	91,0	90,2	88,5
15	90,2	91,0	90,2	89,5
18,5	91,0	92,4	91,7	89,5
22	91,0	92,4	91,7	91,0
30	91,7	93,0	93,0	91,0
37	92,4	93,0	93,0	91,7
45	93,0	93,6	93,6	91,7
55	93,0	94,1	93,6	93,0
75	93,6	94,5	94,1	93,0
90	94,5	94,5	94,1	93,6
110	94,5	95,0	95,0	93,6
150	95,0	95,0	95,0	93,6
185	95,4	95,0	95,0	93,6
220 up to 335	95,4	95,4	95,0	93,6
375 up to 1 000	95,4	95,8	95,0	94,1

5.4.3 Nominal efficiency limits for IE3 (see Tables 7 and 8)

Table 7 – Nominal efficiency limits (%) for 50 Hz IE3

P_N kW	Number of poles/synchronous speed min ⁻¹			
	2/3000	4/1500	6/1000	8/750
0,12	60,8	64,8	57,7	50,7
0,18	65,9	69,9	63,9	58,7
0,20	67,2	71,1	65,4	60,6
0,25	69,7	73,5	68,6	64,1
0,37	73,8	77,3	73,5	69,3
0,40	74,6	78,0	74,4	70,1
0,55	77,8	80,8	77,2	73,0
0,75	80,7	82,5	78,9	75,0
1,1	82,7	84,1	81,0	77,7
1,5	84,2	85,3	82,5	79,7
2,2	85,9	86,7	84,3	81,9
3	87,1	87,7	85,6	83,5
4	88,1	88,6	86,8	84,8
5,5	89,2	89,6	88,0	86,2
7,5	90,1	90,4	89,1	87,3
11	91,2	91,4	90,3	88,6
15	91,9	92,1	91,2	89,6
18,5	92,4	92,6	91,7	90,1
22	92,7	93,0	92,2	90,6
30	93,3	93,6	92,9	91,3
37	93,7	93,9	93,3	91,8
45	94,0	94,2	93,7	92,2
55	94,3	94,6	94,1	92,5
75	94,7	95,0	94,6	93,1
90	95,0	95,2	94,9	93,4
110	95,2	95,4	95,1	93,7
132	95,4	95,6	95,4	94,0
160	95,6	95,8	95,6	94,3
200 up to 1 000	95,8	96,0	95,8	94,6

Table 8 – Nominal efficiency limits (%) for 60 Hz IE3

P_N kW	Number of poles / synchronous speed min ⁻¹			
	2/3600	4/1800	6/1200	8/900
0,12	62,0	66,0	64,0	59,5
0,18	65,6	69,5	67,5	64,0
0,25	69,5	73,4	71,4	68,0
0,37	73,4	78,2	75,3	72,0
0,55	76,8	81,1	81,7	74,0
0,75	77,0	83,5	82,5	75,5
1,1	84,0	86,5	87,5	78,5
1,5	85,5	86,5	88,5	84,0
2,2	86,5	89,5	89,5	85,5
3,7	88,5	89,5	89,5	86,5
5,5	89,5	91,7	91,0	86,5
7,5	90,2	91,7	91,0	89,5
11	91,0	92,4	91,7	89,5
15	91,0	93,0	91,7	90,2
18,5	91,7	93,6	93,0	90,2
22	91,7	93,6	93,0	91,7
30	92,4	94,1	94,1	91,7
37	93,0	94,5	94,1	92,4
45	93,6	95,0	94,5	92,4
55	93,6	95,4	94,5	93,6
75	94,1	95,4	95,0	93,6
90	95,0	95,4	95,0	94,1
110	95,0	95,8	95,8	94,1
150	95,4	96,2	95,8	94,5
185 up to 1 000	95,8	96,2	95,8	95,0

5.4.4 Nominal efficiency limits for IE4 (see Tables 9 and 10)

Table 9 – Nominal efficiency limits (%) for 50 Hz IE4

P_N kW	Number of poles/synchronous speed min^{-1}			
	2/3000	4/1500	6/1000	8/750
0,12	66,5	69,8	64,9	62,3
0,18	70,8	74,7	70,1	67,2
0,20	71,9	75,8	71,4	68,4
0,25	74,3	77,9	74,1	70,8
0,37	78,1	81,1	78,0	74,3
0,40	78,9	81,7	78,7	74,9
0,55	81,5	83,9	80,9	77,0
0,75	83,5	85,7	82,7	78,4
1,1	85,2	87,2	84,5	80,8
1,5	86,5	88,2	85,9	82,6
2,2	88,0	89,5	87,4	84,5
3	89,1	90,4	88,6	85,9
4	90,0	91,1	89,5	87,1
5,5	90,9	91,9	90,5	88,3
7,5	91,7	92,6	91,3	89,3
11	92,6	93,3	92,3	90,4
15	93,3	93,9	92,9	91,2
18,5	93,7	94,2	93,4	91,7
22	94,0	94,5	93,7	92,1
30	94,5	94,9	94,2	92,7
37	94,8	95,2	94,5	93,1
45	95,0	95,4	94,8	93,4
55	95,3	95,7	95,1	93,7
75	95,6	96,0	95,4	94,2
90	95,8	96,1	95,6	94,4
110	96,0	96,3	95,8	94,7
132	96,2	96,4	96,0	94,9
160	96,3	96,6	96,2	95,1
200	96,5	96,7	96,3	95,4
250	96,5	96,7	96,5	95,4
315 up to 1 000	96,5	96,7	96,6	95,4

NOTE Tables 9 and 10 supersede Annex A of IEC 60034-31:2010.

Table 10 – Nominal efficiency limits (%) for 60 Hz IE4

P_N kW	Number of poles / synchronous speed min ⁻¹			
	2/3600	4/1800	6/1200	8/900
0,12	66,0	70,0	68,0	64,0
0,18	70,0	74,0	72,0	68,0
0,25	74,0	77,0	75,5	72,0
0,37	77,0	81,5	78,5	75,5
0,55	80,0	84,0	82,5	77,0
0,75	82,5	85,5	84,0	78,5
1,1	85,5	87,5	88,5	81,5
1,5	86,5	88,5	89,5	85,5
2,2	88,5	91,0	90,2	87,5
3,7	89,5	91,0	90,2	88,5
5,5	90,2	92,4	91,7	88,5
7,5	91,7	92,4	92,4	91,0
11	92,4	93,6	93,0	91,0
15	92,4	94,1	93,0	91,7
18,5	93,0	94,5	94,1	91,7
22	93,0	94,5	94,1	93,0
30	93,6	95,0	95,0	93,0
37	94,1	95,4	95,0	93,6
45	94,5	95,4	95,4	93,6
55	94,5	95,8	95,4	94,5
75	95,0	96,2	95,8	94,5
90	95,4	96,2	95,8	95,0
110	95,4	96,2	96,2	95,0
150	95,8	96,5	96,2	95,4
185	96,2	96,5	96,2	95,4
220	96,2	96,8	96,5	95,4
250 up to 1 000	96,2	96,8	96,5	95,8

NOTE Tables 9 and 10 supersede Annex A of IEC 60034-31:2010.

5.4.5 Interpolation of nominal efficiency limits of intermediate rated powers for 50 Hz mains supply frequency

To determine normative nominal efficiency limits of 50 Hz motors with rated powers not given in the tables above within the range of 0,12 kW up to 200 kW the following formula shall be applied:

$$\eta_N = A \cdot \left[\log_{10} \left(\frac{P_N}{1 \text{ kW}} \right) \right]^3 + B \cdot \left[\log_{10} \left(\frac{P_N}{1 \text{ kW}} \right) \right]^2 + C \cdot \log_{10} \left(\frac{P_N}{1 \text{ kW}} \right) + D$$

where A, B, C, D = interpolation coefficients (see Tables 11 and 12); P_N is given in kW.

NOTE The formula and interpolation coefficients were mathematically derived to create a best fitting curve for the desired nominal efficiency limits. They do not have a physical meaning.

The resulting efficiency (%) shall be rounded to the nearest tenth, i.e. xx,x %.

Table 11 – Interpolation coefficients for 0,12 kW up to 0,74 kW

IE code	Coefficients	8-poles 750 /min	6-poles 1 000 /min	4-poles 1 500 /min	2-poles 3 000 /min
IE1	A	5,9466	–45,9652	16,7271	11,924
	B	7,9458	–87,1474	12,7136	6,3699
	C	40,441	–8,2383	25,947	30,0509
	D	66,146	68,7303	76,174	76,6136
IE2	A	6,4855	–15,9218	17,2751	22,4864
	B	9,4748	–30,258	23,978	27,7603
	C	36,852	16,6861	35,5822	37,8091
	D	70,762	79,1838	84,9935	82,458
IE3	A	–0,5896	–17,361	7,6356	6,8532
	B	–25,526	–44,538	4,8236	6,2006
	C	4,2884	–3,0554	21,0903	25,1317
	D	75,831	79,1318	86,0998	84,0392
IE4	A	–4,9735	–13,0355	8,432	–8,8538
	B	–21,453	–36,9497	2,6888	–20,3352
	C	2,6653	–4,3621	14,6236	8,9002
	D	79,055	82,0009	87,6153	85,0641

Table 12 – Interpolation coefficients for 0,75 kW up to 200 kW

IE code	Coefficients	8-poles 750 /min	6-poles 1 000 /min	4-poles 1 500 /min	2-poles 3 000 /min
IE1	A	2,4433	0,0786	0,5234	0,5234
	B	–13,8	–3,5838	–5,0499	–5,0499
	C	30,656	17,2918	17,4180	17,4180
	D	65,238	72,2383	74,3171	74,3171
IE2	A	2,1311	0,0148	0,0278	0,2972
	B	–12,029	–2,4978	–1,9247	–3,3454
	C	26,719	13,2470	10,4395	13,0651
	D	69,735	77,5603	80,9761	79,077
IE3	A	0,7189	0,1252	0,0773	0,3569
	B	–5,1678	–2,613	–1,8951	–3,3076
	C	15,705	11,9963	9,2984	11,6108
	D	77,074	80,4769	83,7025	82,2503
IE4	A	0,6556	0,3598	0,2412	0,34
	B	–4,7229	–3,2107	–2,3608	–3,0479
	C	13,977	10,7933	8,446	10,293
	D	80,247	84,107	86,8321	84,8208

5.4.6 Interpolation of nominal efficiency limits of intermediate rated powers for 60 Hz mains supply frequency

Normative, nominal limits are given in Tables 4, 6, 8 and 10. Normative, nominal limits of 60 Hz motors having rated powers not defined in the tables shall be determined as follows:

- The efficiency of a rated power at or above the midpoint between two consecutive power values from the tables shall be the higher of the two efficiencies.
- The efficiency of a rated power below the midpoint between two consecutive power values from the tables shall be the lower of the two efficiencies.

Annex A (informative)

Nominal limits for efficiency class IE5

The levels of the IE5 efficiency class are envisaged to be incorporated into the next edition of this standard and/or in part two of this standard series. It is the goal to reduce the losses of IE5 by some 20 % relative to IE4. Motor technologies for IE5 are currently not well developed and not commercially available.

Further energy-efficiency optimizations will have to focus on improved system efficiency throughout the entire operating load cycle including all system-losses (converter, filter, cables, motor, etc.), see EN 52800.

Bibliography

IEC 60034-5, *Rotating electrical machines – Part 5: Degrees of protection provided by the integral design of rotating electrical machines (IP code) – Classification*

IEC 60034-12, *Rotating electrical machines – Part 12: Starting performance of single-speed three-phase cage induction motors*

IEC/TS 60034-31:2010, *Rotating electrical machines – Part 31: Selection of energy-efficient motors including variable speed applications – Application guide*

IEC 60072-1, *Dimensions and output series for rotating electrical machines – Part 1: Frame numbers 56 to 400 and flange numbers 55 to 1080*

ISO 3, *Preferred numbers – Series of preferred numbers*

EN 12101-3, *Smoke and heat control systems – Part 3: Specification for powered smoke and heat exhaust ventilators*

EN 50347, *General purpose three-phase induction motors having standard dimensions and outputs – Frame numbers 56 to 315 and flange numbers 65 to 740*

EN 52800, *Energy efficiency for power drive systems, motor starters, power electronics and their driven applications*

JIS C 4212 (Japanese Industrial Standard), *Low-voltage three-phase squirrel-cage high-efficiency induction motors*

NBR 17094-1, *Rotating electrical machines – Induction motors – Specification*

NEMA MG1, *Motors and Generators*

SANS 1804-1 (South African Standard), *Induction motors – Part 1: IEC requirements*
